

COMMON COURSE ASSIGNMENT FORMAT

A. Assignment Information

Particular	Details
Department:	COMPUTER SCIENCE
Programme:	ARTIFICIAL INTELLIGENCE
Course Code & Course Name	CSA1703
Academic Year / Batch	2026/2025
Faculty Name	DR JEENA R
Assignment Title	Automated Sign Language Recognition and Speech Conversion System
Date of Issue	02-09-2026
Date of Submission	02-09-2026
Maximum Marks	100
Course Outcome(s) – CO	CO1, CO2, CO3, CO4, CO5
Bloom's Taxonomy Level	L4 – Analyze / L5 – Evaluate / L6 – Create
SDG Mapping (SDG 1–17)	SDG 3 – Good Health and Well-Being; SDG 4 – Quality Education; SDG 10 – Reduced Inequalities; SDG 9 – Industry, Innovation and Infrastructure
Industry / Societal Relevance	The system provides an AI-based assistive communication solution for deaf and hard-of-hearing individuals. It has applications in healthcare, education, banking, public services, customer support, and other accessibility-focused environments. It demonstrates the practical use of computer vision, machine learning, deep learning, and speech technologies to improve inclusive communication.

A. Assignment Problem / Challenge

Communication between deaf or hard-of-hearing individuals who use sign language and people who do not understand sign language can be difficult. In many public places such as hospitals, educational institutions, banks, railway stations, and government offices, the absence of a common communication medium can create communication barriers.

The challenge is to design and develop an **Automated Sign Language Recognition and Speech Conversion System** using computer vision, image processing, machine learning/deep learning, and speech technologies. The system should capture hand gestures through a camera, identify the corresponding sign, convert the recognized sign into text, and generate spoken audio from the resulting text.

The assignment requires students to:

- Analyze the challenges involved in recognizing human hand gestures.
- Collect or use an appropriate sign-language image/video dataset.
- Perform image preprocessing and feature extraction.
- Develop a machine-learning/deep-learning-based sign recognition model.
- Convert recognized signs into meaningful text.
- Implement text-to-speech conversion.
- Evaluate the system using suitable performance metrics.
- Compare alternative recognition approaches where appropriate.
- Analyze technical, ethical, accessibility, and societal considerations.

B. Problem Statement

Problem / Case / Design Challenge

Develop an **Automated Sign Language Recognition and Speech Conversion System** that uses a camera to capture hand gestures and automatically recognizes predefined sign-language gestures.

The proposed system should process the captured image or video, identify the hand gesture using computer vision and machine-learning/deep-learning techniques, map the recognized gesture to its corresponding word or character, and convert the generated text into speech.

The proposed system follows the workflow:

Camera Input → Image/Video Acquisition → Preprocessing → Hand Detection → Feature Extraction → Sign Classification → Text Generation → Text-to-Speech → Audio Output

The system should demonstrate that automated sign recognition can provide a practical communication interface between sign-language users and non-sign-language users.

C. Requirements and Constraints

Students shall identify and consider the following requirements and constraints relevant to the proposed system.

1. Functional Requirements

The system shall:

1. Capture hand gestures using a webcam or camera.
2. Detect the hand or gesture from the input image or video.
3. Preprocess the captured input.
4. Extract relevant visual features.
5. Classify the gesture into a predefined sign category.
6. Display the recognized sign as text.
7. Convert the recognized text into speech.
8. Provide the speech through a speaker or audio output.
9. Display recognition confidence where possible.
10. Provide real-time or near-real-time recognition.

2. Performance Requirements

The system should aim to achieve:

- High gesture-classification accuracy.
- Low prediction latency.
- Reliable recognition under normal lighting conditions.
- Robustness to moderate changes in hand position and orientation.
- Reduced false recognition.
- Smooth conversion from recognized text to speech.

3. Technical Constraints

The following technical constraints should be considered:

- A webcam or camera is required for live recognition.
- The system may be developed using Python.
- Training requires a suitable dataset containing labeled sign-language images or videos.
- Computational resources may be limited during deep-learning model training.
- Recognition accuracy may decrease under poor lighting, occlusion, complex backgrounds, or unusual hand orientations.
- The initial prototype may be restricted to a predefined vocabulary or alphabet.

4. Cost and Resource Constraints

The proposed system should preferably use:

- Open-source software.
- Freely available datasets where licensing permits.
- Standard computer hardware.

- A webcam instead of specialized sensing equipment.
- Free or open-source machine-learning libraries.

5. Safety and Reliability

- The system should not be considered a complete replacement for professional human sign-language interpreters in critical situations.
- Recognition results should be treated as an assistive communication mechanism.
- The system should indicate uncertainty when prediction confidence is low.

6. Privacy and Security

- Camera data should be processed locally wherever practical.
- Images and videos containing people should not be stored unnecessarily.
- Personal information should not be collected without permission.
- Datasets must be used according to their licenses and applicable privacy requirements.

7. Ethical and Accessibility Considerations

- The system should support accessibility and inclusive communication.
- Training data should contain sufficient variation in users, hand sizes, skin tones, lighting conditions, and signing styles.
- Users should be informed about the limitations of automated recognition.
- The system should not claim complete sign-language understanding if it recognizes only a limited vocabulary.

D. Student Work

1. Problem Understanding and Formulation

What is the Problem?

The main problem is the communication gap between people who use sign language and people who do not understand it. Traditional communication may require the assistance of another person who understands sign language.

The proposed system attempts to reduce this communication barrier by automatically recognizing predefined hand gestures and converting them into text and speech.

Expected Outcomes

The expected system should provide the following:

Input:

Hand gesture captured using a camera.

Processing:

Image preprocessing → Hand detection → Feature extraction → Classification.

Output:

Recognized sign → Text → Speech.

Available Data

The project may use:

- Sign-language image datasets.
- Sign-language video datasets.
- Self-collected images.
- Webcam-generated test images.
- Labeled hand-gesture samples.

Each sample should preferably contain a corresponding class label.

Assumptions

The following assumptions may be made:

- The camera provides sufficiently clear images.
- The user performs predefined signs correctly.
- The initial model recognizes a limited number of signs.
- The system operates under reasonable lighting conditions.
- The selected dataset contains sufficient examples for training and testing.

Constraints

Important constraints include:

- Limited vocabulary.
- Background variation.
- Lighting variation.
- Hand occlusion.
- Similar-looking gestures.
- Camera resolution.
- Processing speed.
- Training-data availability.
- Computational resources.

2. Application of Course Knowledge

The project applies concepts from **Artificial Intelligence, Machine Learning, Computer Vision, Image Processing, Pattern Recognition, and Speech Processing.**

A. Image Processing

The captured image can be processed using:

- Image resizing.
- Noise removal.
- Color conversion.
- Normalization.
- Background reduction.
- Image enhancement.

The preprocessing stage improves the quality and consistency of the input before recognition.

B. Hand Detection

A hand-detection technique can be used to identify the region containing the hand.

Possible approaches include:

- OpenCV-based segmentation.
- Hand landmark detection.
- Background subtraction.
- Deep-learning-based object detection.

C. Feature Extraction

Important gesture features may include:

- Hand shape.
- Finger positions.
- Finger angles.
- Palm orientation.
- Hand landmarks.
- Relative distances between landmarks.

D. Classification

Possible classification algorithms include:

- K-Nearest Neighbors (KNN).
- Support Vector Machine (SVM).
- Random Forest.
- Decision Tree.
- Artificial Neural Network (ANN).
- Convolutional Neural Network (CNN).

For image-based recognition, a CNN can learn spatial features directly from training images.

E. Text Generation

After classification, the predicted class is mapped to the corresponding sign or character.

Predicted Class → Corresponding Sign/Character → Text

For example:

Gesture → “HELLO” → Display “HELLO”

F. Speech Conversion

The generated text is passed to a text-to-speech module.

Recognized Text → Text-to-Speech Engine → Audio Output

3. Solution / Design / Methodology

Proposed System Architecture

The proposed system consists of the following major modules:

1. Camera Input Module
2. Image Acquisition Module
3. Image Preprocessing Module
4. Hand Detection Module
5. Feature Extraction Module
6. Sign Classification Module
7. Text Generation Module
8. Text-to-Speech Module
9. User Interface Module

Step 1: Input Acquisition

A webcam captures the user's hand gesture in real time.

Step 2: Preprocessing

The captured image is:

1. Resized.
2. Converted into an appropriate color representation.
3. Normalized.
4. Processed to reduce noise.
5. Prepared for recognition.

Step 3: Hand Detection

The system identifies the hand region and extracts relevant hand information.

Step 4: Feature Extraction

The system extracts visual or landmark-based features from the detected hand.

Step 5: Sign Classification

The extracted features are provided to a trained machine-learning or deep-learning model.

Example:

Input Gesture	Predicted Sign
Gesture 1	A
Gesture 2	B
Gesture 3	C
Gesture 4	HELLO
Gesture 5	THANK YOU

Step 6: Text Generation

The recognized sign is converted into a text representation.

For example:

Gesture 1 → H

Gesture 2 → E

Gesture 3 → L

Gesture 4 → L

Gesture 5 → O

The resulting text is:

HELLO

Step 7: Speech Conversion

The generated text is sent to a text-to-speech engine.

“HELLO” → Audio Output

Step 8: User Interface

The interface can display:

- Camera feed.
- Detected hand.
- Recognized sign.
- Confidence score.
- Generated text.
- Speech status.

Alternative Solutions

Students can compare two or more approaches.

Approach	Advantages	Limitations
Traditional Image Processing + ML	Simple and lightweight	Sensitive to lighting and background
Hand Landmarks + ML	Efficient and requires fewer image features	Depends on accurate landmark detection
CNN	Strong image-recognition capability	Requires more training resources
CNN + Hand Detection	Better focus on hand region	More computationally complex

Selected Approach

For the proposed prototype, a **CNN-based image classification approach or hand-landmark-based machine-learning approach** may be selected depending on the available dataset and computational resources.

The selected approach should be justified based on:

- Accuracy.
- Processing speed.
- Dataset size.
- Computational requirements.

4. Use of Modern Tools

The following modern tools can be used for implementing the system.

Programming Language

Python

Computer Vision

- OpenCV
- MediaPipe or another suitable hand-landmark framework

Machine Learning

- Scikit-learn

Deep Learning

- TensorFlow/Keras or PyTorch

Data Processing

- NumPy
- Pandas

Data Visualization

- Matplotlib
- Plotly

Speech Processing

- Python text-to-speech library or API

Development Environment

- Jupyter Notebook
- Google Colab

- Visual Studio Code
- Python IDLE

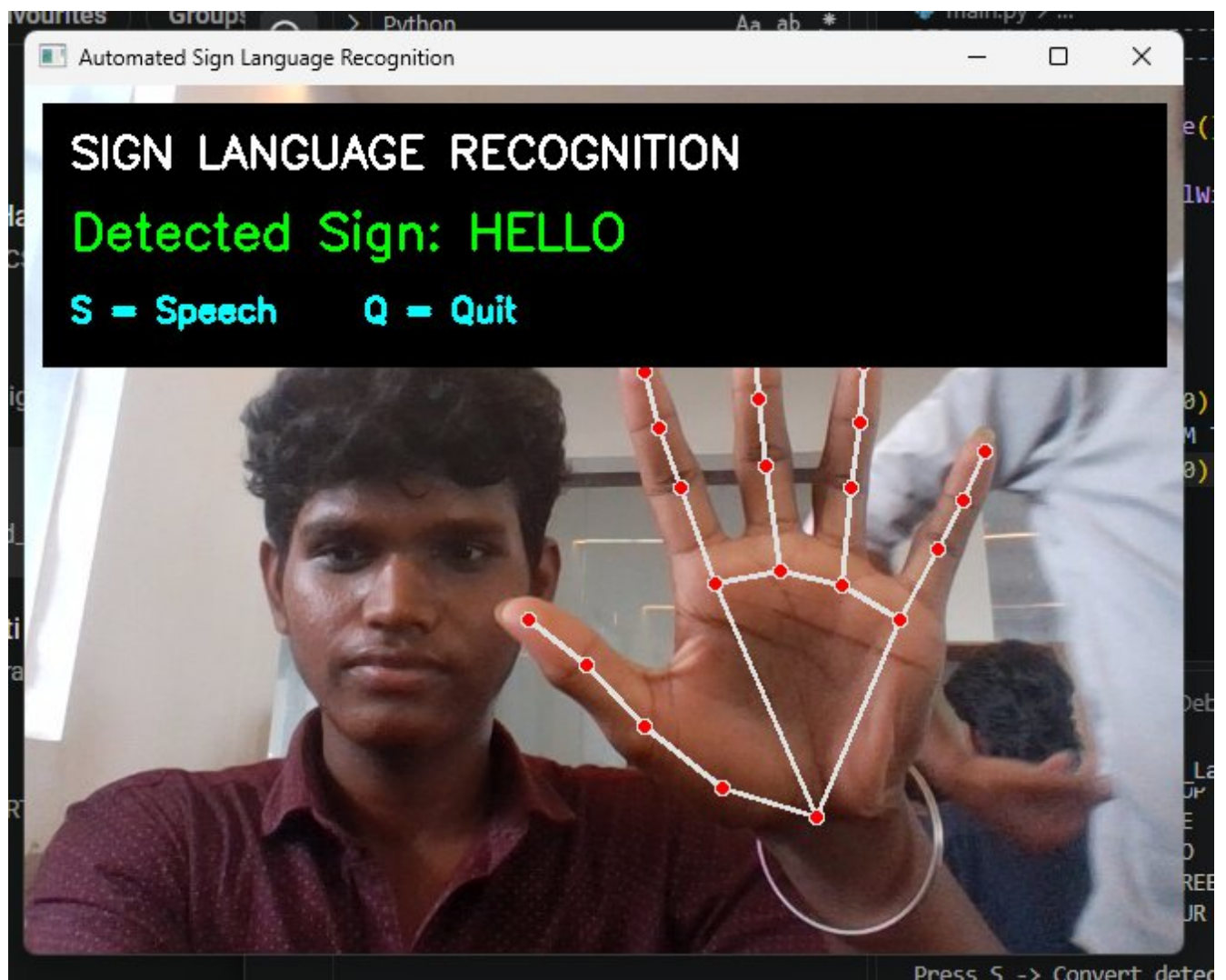
Version Control

- Git
- GitHub

Students should provide evidence of tool usage, such as:

- Dataset screenshots.
- Training outputs.
- Accuracy and loss graphs.
- Confusion matrix.
- Prediction screenshots.
- Live webcam recognition.
- Text-to-speech output.

5. Results and Validation



The developed system should be evaluated using appropriate performance measures.

A. Accuracy

Accuracy measures the proportion of correctly classified samples.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

For multiclass sign recognition:

$$\text{Accuracy} = \text{Number of Correct Predictions} / \text{Total Number of Predictions}$$

B. Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Precision measures the proportion of predicted positive samples that are actually correct.

C. Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Recall measures the ability of the system to correctly identify relevant samples.

D. F1-Score

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

The F1-score provides a balance between precision and recall.

E. Confusion Matrix

A confusion matrix should be generated to determine which signs are correctly and incorrectly classified.

Suggested Results Table

Metric	Result
Training Accuracy	_____ %
Validation Accuracy	_____ %
Testing Accuracy	_____ %
Precision	_____

Recall	_____
F1-Score	_____
Average Prediction Time	_____ms

Additional Testing

The system should be tested under different conditions.

Test Condition	Expected Observation
Good lighting	High recognition accuracy
Low lighting	Accuracy may decrease
Plain background	Better recognition
Complex background	Possible false detection
Different hand positions	Moderate variation
Different users	Measures generalization
Fast hand movement	Possible recognition errors

The results should be analyzed to determine whether the proposed system satisfies the specified functional and performance requirements.

6. Analysis and Engineering Decision

The experimental results should be analyzed to determine the effectiveness of the proposed system.

Students should compare the selected model with alternative approaches based on measurable factors such as accuracy, prediction time, computational requirements, and implementation complexity.

Example Comparison

Parameter	Traditional ML	CNN
Training Complexity	Low	High
Feature Extraction	Usually Manual	Learned Automatically
Image Recognition	Moderate	Generally Strong
Computational Requirement	Low	Higher
Dataset Requirement	Moderate	Usually Higher
Real-Time Potential	High	Depends on Model

Engineering Decision

The final model should be selected based on experimental results rather than theoretical advantages alone.

For example, if a CNN provides significantly better recognition accuracy while maintaining acceptable prediction speed, it may be selected as the final solution.

However, if computational resources are limited and a landmark-based classifier provides sufficiently high accuracy, the lighter approach may be preferred.

The final decision should be supported by quantitative or qualitative evidence obtained during testing.

7. Broader Considerations

A. Social Impact

The proposed system can help reduce communication barriers for people who use sign language.

Potential application areas include:

- Hospitals.
- Schools and colleges.
- Public offices.
- Banks.
- Transportation facilities.
- Customer-service environments.

B. Accessibility

The system can provide an additional communication channel without requiring a human interpreter for every basic interaction.

C. Ethical Considerations

The system should not claim perfect understanding of sign language. Recognition errors may result in incorrect communication, particularly in critical situations.

D. Privacy

Camera-based systems may capture sensitive visual information. Therefore:

- Unnecessary video storage should be avoided.
- Data should be processed securely.
- User consent should be obtained when collecting data.

E. Economic Considerations

Using a webcam and open-source software can reduce development costs compared with specialized hardware-based solutions.

F. Sustainability

A software-based solution using commonly available computing devices can reduce the need for dedicated hardware. Efficient machine-learning models can also reduce computational and energy requirements.

G. Professional Responsibility

Developers must clearly communicate:

- System capabilities.
- System limitations.
- Dataset limitations.
- Recognition accuracy.

- Potential risks of incorrect predictions.

8. Conclusion

The **Automated Sign Language Recognition and Speech Conversion System** provides an AI-based approach for reducing communication barriers between sign-language users and people who do not understand sign language.

The proposed system captures hand gestures through a camera, preprocesses the input, detects relevant hand information, extracts features, classifies the gesture using machine learning or deep learning, converts the recognized sign into text, and finally generates speech using a text-to-speech mechanism.

The effectiveness of the system can be evaluated using accuracy, precision, recall, F1-score, confusion matrix, and prediction time. Although challenges such as lighting variation, background complexity, gesture similarity, limited vocabulary, and user variation may affect performance, the system provides a useful foundation for developing accessible human-computer communication technologies.

Future improvements can include:

- Recognition of continuous sign-language sentences.
- Larger sign-language vocabulary.
- Two-way communication.
- Mobile application deployment.

9. Student Reflection

Student 1

This assignment helped me understand how computer vision and machine learning can be applied to solve a real-world accessibility problem. I learned how image preprocessing, feature extraction, classification, and model evaluation work together in an AI application.

If additional time and resources were available, I would improve the system by increasing the dataset size and testing it with more users and environmental conditions.

Student 2

Through this assignment, I learned that developing an AI system involves more than training a machine-learning model. Dataset quality, preprocessing, feature selection, testing, and evaluation significantly affect the final performance. I also learned about the importance of accessibility and ethical considerations in AI applications.

With additional resources, I would implement continuous sign-language recognition instead of recognizing only individual signs.

Student 3

I learned how computer vision can be combined with speech technology to create an interactive application. The project improved my understanding of real-time image processing and machine-learning model deployment.

I would further improve the system by implementing a mobile application and supporting multiple languages.

Student 4

This assignment helped me understand the practical challenges of deploying an AI model in real-world environments. I learned about recognition errors, lighting conditions, background variation, and model generalization.

If given additional resources, I would develop a larger and more diverse dataset to improve recognition reliability.

10. References

The following types of sources should be acknowledged in the final project:

1. Research papers related to automated sign-language recognition.
2. Sign-language datasets used for training and testing.

3. OpenCV documentation.
4. TensorFlow/Keras or PyTorch documentation.
5. Scikit-learn documentation.
6. Documentation for the selected hand-detection or hand-landmark framework.
7. Documentation for the selected text-to-speech technology.
8. External images, datasets, libraries, APIs, and software used in the project.
9. AI-assisted tools used during the development or preparation of the assignment.

B. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

C. CO–PO–Assessment Mapping

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation	CO5	PO1,PO2	L3/L4	10
Application of knowledge	CO5	PO1,PO3	L3/L4	20
Solution / Design / Implementation	CO5	PO3,PO5	L4/L5/L6	20
Modern tool usage	CO5	PO5	L3/L4	10
Validation	CO5	PO4,PO5	L4/L5	15
Analysis & justification	CO5	PO2,PO4	L4/L5	15
Broader considerations	CO5	PO6,PO7	L4/L5	5
Documentation & reflection	CO5	PO9,PO10	L3/L4	5

Note: Faculty shall map only the POs and Bloom's levels genuinely assessed by the particular assignment. Every assignment is **not required to map to every PO**.

D. Faculty Evaluation & Continuous Improvement

CO Attainment

Performance Level	Criterion
Level 3	$\geq 70\%$
Level 2	60–69%
Level 1	50–59%
Level 0	$< 50\%$

Target CO Attainment: $\geq 70\%$

Actual CO Attainment: 90%

Faculty Analysis

- **Areas in which students performed well: Students demonstrated good understanding of the topic.**
- **Areas requiring improvement: Some students lacked depth in their explanations**
- **Corrective / Improvement Action: Provided additional guidance and examples.**
- **Documents to be Maintained: Include more topic-wise examples in the next offering.**

The department/course faculty shall maintain:

1. Assignment question/problem statement
2. CO–PO and Bloom's mapping
3. Assessment rubric
4. Student submissions
5. Tool/simulation/experimental evidence, where applicable
6. Evaluated scripts/reports
7. Marks and rubric-wise assessment
8. CO attainment analysis
9. Sample student work representing different performance levels
10. Faculty analysis and continuous-improvement action

